ASSIGNMENT-2 SOLUTIONS: Applied Concepts

COURSE	SUBJECT CODE	SUBJECT TITLE	SUBMISSION DATE
BCA	BCA309T-AS02	Foundations of Cryptography	08 October 2026

Q1. List the four fundamental cryptographic goals/objectives. SHORT ANSWER

Answer: The four fundamental goals of modern cryptography are:

- **Confidentiality:** Protecting sensitive data from unauthorized interception or disclosure.
- **Integrity:** Ensuring data is not altered, modified, or corrupted during transmission or storage.
- **Authentication:** Verifying the genuine identity of communicating parties or data sources.
- **Non-repudiation:** Preventing a party from falsely denying the authenticity of a sent message or transaction.

Q2. Define "Confidentiality" as a cryptographic goal with one example. SHORT ANSWER

Answer: Confidentiality guarantees that information remains secret and accessible only to authorized entities holding the appropriate decryption keys. It renders transmitted data completely unintelligible to unauthorized eavesdroppers.

Real-World Example: When you enter your password on a website, the protocol encrypts the login credentials using TLS/SSL before sending them across the internet. If an attacker intercepts the packet, they only see scrambled ciphertext instead of your plain password.

Q3. Explain each of the cryptographic objectives in detail with real-world scenarios: Confidentiality, Integrity, Authentication, and Non-repudiation. Why is it important to achieve all four together? LONG ANSWER

Answer:

- **1. Confidentiality:** Ensures sensitive details are concealed from eavesdroppers.
  Scenario: A patient viewing medical records online relies on AES encryption so that unauthorized third parties cannot read their private health diagnosis during network transmission.
- **2. Integrity:** Guarantees that data cannot be modified in transit without detection.
  Scenario: A bank client transfers $100. Cryptographic hash functions (like SHA-256) ensure a attacker cannot alter the amount to $10,000 mid-transit without invalidating the hash verification.
- **3. Authentication:** Validates the authentic identity of the sender or system.
  Scenario: An online banking app uses digital certificates (X.509) to assure the user they are connected to their official bank server, preventing man-in-the-middle phishing attacks.
- **4. Non-repudiation:** Provides binding proof that a specific sender initiated an action.
  Scenario: An executive signs a corporate contract digitally using their private key. Because only they possess that key, they cannot later claim someone else signed the document.

Importance of Combined Security: Securing a system requires achieving all four goals simultaneously. For instance, if a system provides confidentiality without integrity, an attacker could tamper with encrypted messages without being detected. If it provides integrity without authentication, anyone could forge valid-looking tampered messages. All four objectives work together to build total trust in modern digital infrastructures.

Q4. What is the difference between a substitution cipher and a transposition cipher? Give one example of each. SHORT ANSWER

Answer: A **substitution cipher** replaces characters in the plaintext with other letters, numbers, or symbols based on a set rule. In contrast, a **transposition cipher** keeps the original characters intact but rearranges their positions or order to scramble the message.

- *Substitution Example:* **Caesar Cipher** (shifting each letter by 3 positions: CAT → FDW).
- *Transposition Example:* **Rail Fence Cipher** (writing characters diagonally across rows and reading row-by-row: HELLO → HLOEL).

Q5. Discuss the evolution from classical ciphers to modern cryptographic standards (DES → AES, RSA, SHA). Highlight key milestones and motivations for change. LONG ANSWER

Answer:

The progression of cryptography reflects an ongoing race between security designers and rising computing power. Early classical ciphers (e.g., Caesar, Vigenère) relied on manual character manipulations that were easily broken using frequency analysis once computer technology emerged.

Key Milestones and Motivations for Change:

- **DES (Data Encryption Standard - 1977):** Introduced as a standardized symmetric 56-bit key block cipher for computer data. *Motivation for Change:* Advances in processing speed made 56-bit keys vulnerable to brute-force attacks by the late 1990s (cracked in under 22 hours).
- **AES (Advanced Encryption Standard - 2001):** Developed to replace DES. AES supports key sizes of 128, 192, and 256 bits. It is computationally secure against all known brute-force attacks and remains the global standard today.
- **RSA (Public-Key Revolution - 1977):** Introduced asymmetric cryptography based on prime factorization. *Motivation:* Solved the critical key-distribution challenge of symmetric systems by allowing public key sharing.
- **SHA (Secure Hash Algorithms):** Developed by NIST for message integrity. Evolved from SHA-1 (now deprecated due to collision risks) to SHA-2 and SHA-3 to ensure collision-resistant hashing for modern security.

Q6. What is a "key" in cryptography, and why is key management important? SHORT ANSWER

Answer: A **key** is a string of random bits used by a cryptographic algorithm to lock (encrypt) or unlock (decrypt) data, or to create digital signatures. **Key management** is crucial because Kerckhoffs's principle dictates that a cryptographic system's security depends entirely on keeping the key secret, rather than the algorithm itself. Poor key generation, storage, or distribution completely compromises system security.

Q7. Explain, with suitable examples, how cryptographic techniques are used to protect data during transmission, prevent unauthorized access, and detect tampering in a network. LONG ANSWER

Answer: Cryptography safeguards networked communications across three core operational areas:

1. **Protecting Data During Transmission (Confidentiality):** Data traversing public networks is vulnerable to packet sniffing. Protocols like **TLS/SSL** and **IPSec VPNs** encrypt payloads using symmetric ciphers (e.g., AES-GCM). Intercepted packets appear as unreadable random noise to unauthorized entities.
2. **Preventing Unauthorized Access (Authentication & Access Control):** Networks verify entity identities before granting resource access. Public Key Infrastructure (PKI) uses **X.509 Digital Certificates** and protocols like **SSH** (Secure Shell) to authenticate servers and users via asymmetric key pairs, preventing unauthorized connections.
3. **Detecting Tampering (Integrity Verification):** Routers and receivers verify that transmitted packets have not been modified. Sender systems generate a **Cryptographic Message Authentication Code (HMAC)** or digital hash (e.g., SHA-256) attached to the data. If an attacker modifies even a single bit in transit, the recalculated hash fails to match, alerting the system to discard the packet.

Q8. Differentiate between a passive attack and an active attack on a cryptographic system, with one example of each. SHORT ANSWER

Answer: A **passive attack** attempts to learn or make use of information from the system without affecting system resources or modifying data (focused on monitoring). An **active attack** involves altering system resources, modifying data packets, or forging communications to disrupt operations.

- *Passive Attack Example:* **Packet Sniffing / Eavesdropping** (listening to unencrypted Wi-Fi traffic).
- *Active Attack Example:* **Man-in-the-Middle (MITM) / Message Modification** (intercepting and altering a bank wire request).

Q9. Describe how cryptographic techniques evolved historically to counter emerging security threats, highlighting the key-distribution problem. LONG ANSWER (PART 1)

Answer: Cryptography evolved continuously as new threats challenged existing protection methods. In purely **symmetric cryptographic systems**, both communicating parties must share the exact same secret key. This created the fundamental **Key Distribution Problem**: how can two parties securely exchange a secret key over an insecure, public network without an eavesdropper intercepting it?

(Q9 Continued) To solve the key-distribution bottleneck, Whitfield Diffie and Martin Hellman introduced **Asymmetric (Public-Key) Cryptography** in 1976. By using mathematically linked key pairs (a Public Key for encryption and a Private Key for decryption), sender and receiver no longer need to share secret keys beforehand. Anyone can encrypt a message using the recipient's public key, but only the holder of the matching private key can decrypt it. This innovation eliminated the requirement for secure physical key exchanges and enabled large-scale internet security.

Q10. Give one example each of a modern symmetric algorithm and a modern asymmetric algorithm used today. SHORT ANSWER

Answer:

- **Modern Symmetric Algorithm: AES (Advanced Encryption Standard)** — widely used in disk encryption (BitLocker), wireless security (WPA3), and TLS web encryption.
- **Modern Asymmetric Algorithm: RSA (Rivest–Shamir–Adleman)** or **ECC (Elliptic Curve Cryptography)** — extensively used for digital signatures, secure key exchanges, and SSL/TLS certificates.

Q11. Discuss the classification of cryptography into Classical Cryptography and Modern Cryptography. Explain the transition from manual/mechanical methods to computer-based algorithms. LONG ANSWER

Answer:

Cryptography is broadly classified into two distinct eras based on techniques, mathematical foundation, and operational medium:

Dimension	Classical Cryptography	Modern Cryptography
Operating Basis	Character-based (alphabets, letters).	Bit-based (binary 0s and 1s).
Methods	Manual paper-and-pencil or simple rotor machines.	Computer algorithms & mathematical functions.
Security Principle	Security through key secrecy & obscure rules.	Kerckhoffs's principle (algorithm public, key secret).
Primary Focus	Confidentiality in military/diplomatic text.	Confidentiality, Integrity, Authentication, Non-repudiation.

Transition Milestones:

- **Manual Stage:** Hand-written substitution and transposition ciphers (e.g., Caesar, Playfair).
- **Mechanical / Electromechanical Era:** Early 20th-century rotor machines (e.g., Germany's Enigma, M-209). Decrypted using early computing engines like Turing's *Bombe*.
- **Computer-Based Era:** Shifted to binary bit manipulation (XOR operations, substitution-permutation networks). Standardized algorithms like DES, AES, RSA, and SHA now handle high-speed automated digital encryption across global computer networks.